

AMIT CHAUHAN

B.Tech Student in Computer Science and Engineering

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Summary

Computer Science undergraduate with hands-on experience in Python, Machine Learning, NLP, Computer Vision, and full-stack application development. Built AI-powered applications using YOLOv8, OpenCV, Flutter, and SQLite while strengthening analytical and problem-solving skills through Data Structures & Algorithms and Object-Oriented Programming. Passionate about Software Development, AI/ML, and building scalable real-world solutions.

Technical Skills

- **Programming Languages:** C, Python
- **Core Concepts:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Problem Solving
- **AI/ML:** Machine Learning, Computer Vision, Deep Learning, NLP
- **Libraries & Frameworks:** NumPy, Pandas, Scikit-learn, FastAPI
- **Databases:** MySQL, SQLite
- **Tools & Platforms:** Git, GitHub, VS Code, Jupyter Notebook

Projects

Podcast Recommendation System | *Python, Machine Learning, NLP, TF-IDF, SQLite, Flutter*

- Built a full-stack podcast recommendation platform using Flutter frontend and Python backend.
- Designed a TF-IDF-based recommendation engine using user interaction and search history data for personalized content delivery.
- Integrated YouTube Data API for real-time podcast streaming and metadata retrieval.
- Optimized SQLite database schema for efficient data storage and faster query execution.
- Enhanced content discovery using NLP-based recommendation techniques.
- Enhanced personalized recommendations using user behavior analysis.

AI-Based Object Detection System for Visually Impaired Users | *YOLOv8, OpenCV, Deep Learning, TTS, Flutter*

- Created an AI-powered object detection system for visually impaired users using YOLOv8 and OpenCV.
- Implemented real-time object recognition and voice alert functionality using Text-to-Speech (TTS).
- Improved object detection accuracy and real-time performance through custom dataset training and testing.
- Applied deep learning and computer vision techniques for environmental awareness and accessibility assistance.
- Improved accessibility for visually impaired users through voice-assisted object detection.

Education

- **GLS University, Ahmedabad** | Bachelor of Technology (B.Tech) | 2023 – 2027 | **SGPA: 7.94 (CURRENT)**
- **Royal Science School, Dhoraji** | 12th Science - HSC Board | 2021 – 2023 | **Percentage: 60.5%**
- **Shree Sarasvati Vidhyalaya, Jam Kandorna** | 10th - GSEB Board | 2020 – 2021 | **Percentage: 91.5%**

Certifications

- Introduction to Programming Using Python – *Udemy*
- Data Structures and Algorithms: The Complete Bootcamp – *Udemy*
- Python Machine Learning: From Beginner to Pro – *Udemy*

Languages

- Gujarati (Native), Hindi (Fluent), English (Proficient)